Homework 7

Solutions

Exercise 7.1 (Panaretos, Exercise 32)**.** Let X be a discrete random variable taking the values

$$\begin{cases} 0 & \text{with probability } 6\theta^2 - 4\theta + 1 \\ 1 & \text{with probability } \theta - 2\theta^2 \\ 2 & \text{with probability } 3\theta - 4\theta^2, \end{cases}$$

where $\theta \in \left[0, \frac{1}{2}\right]$. Determine the maximum likelihood estimator on the basis of a sample of size 1, X_1. What do you observe?

Sol. The likelihood function for a single sample is given by

$$L(\theta) = \begin{cases} 6\theta^2 - 4\theta + 1 & \text{if } X_1 = 0 \\ \theta - 2\theta^2 & \text{if } X_1 = 1 \\ 3\theta - 4\theta^2 & \text{if } X_1 = 2. \end{cases}$$

Instead of taking the logarithm, we just maximize the likelihood directly in each case. To find critical points, we notice that

$$L'(\theta) = \begin{cases} 12\theta - 4 & \text{if } X_1 = 0 \\ 1 - 4\theta & \text{if } X_1 = 1 \\ 3 - 8\theta & \text{if } X_1 = 2. \end{cases}$$

with critical point

$$\hat{\theta} = \begin{cases} 1/3 & \text{if } X_1 = 0 \\ 1/4 & \text{if } X_1 = 1 \\ 3/8 & \text{if } X_1 = 2. \end{cases}$$

However, this is **not** the MLE. Indeed, the second derivative is given by

$$L''(\theta) = \begin{cases} 12 & \text{if } X_1 = 0 \\ -4 & \text{if } X_1 = 1 \\ -8 & \text{if } X_1 = 2 \end{cases}$$

and so this critical point is only a (local) maximum in the $X_1 = 1, 2$ case. We need to check the endpoints, $\theta = 0$ and $\theta = \frac{1}{2}$:

$$L(0) = \begin{cases} 1 & \text{if } X_1 = 0 \\ 0 & \text{if } X_1 = 1 \\ 0 & \text{if } X_1 = 2. \end{cases}, \qquad L\left(\frac{1}{2}\right) = \begin{cases} 1/2 & \text{if } X_1 = 0 \\ 0 & \text{if } X_1 = 1 \\ 1/2 & \text{if } X_1 = 2. \end{cases}$$

Comparing to $\hat{\theta}$,

$$L\left(\hat{\theta}\right) = \begin{cases} 1/3 & \text{if } X_1 = 0 \\ 1/8 & \text{if } X_1 = 1 \\ 9/16 & \text{if } X_1 = 2. \end{cases}$$

In the $X_1 = 0$ case, the maximum is attained at the endpoint $\theta = 0$, and $\hat{\theta}$ was actually a minimum. So, the correct MLE is

$$\boxed{\hat{\theta}_{\mathrm{MLE}} = \begin{cases} 0 & \text{if } X_1 = 0 \\ 1/4 & \text{if } X_1 = 1 \\ 3/8 & \text{if } X_1 = 2. \end{cases}}$$

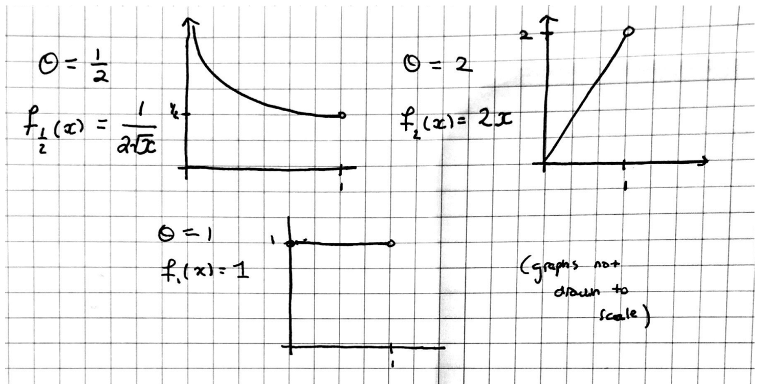
Exercise 7.2 (HTZ, Exercise 6.4.7)**.** Let $X_1, \ldots, X_n$ be sampled i.i.d. from the distribution with probability density function

$$f_\theta(x) = \theta x^{\theta-1}, \qquad 0 < x < 1$$

where θ is an unknown parameter with possible values $0 < \theta < \infty$.

1. Sketch the probability density function of X for $\theta = \frac{1}{2}$, $\theta = 1$ and $\theta = 2$.

   Sol.



2. Show that

$$\hat{\theta} = \frac{-n}{\sum_{i=1}^n \log(X_i)}$$

   is the maximum likelihood estimator for θ.

   Proof. The likelihood function is given by

$$L(\theta) = \prod_{i=1}^n f_\theta(X_i) = \prod_{i=1}^n \theta X_i^{\theta-1} = \theta^n \left(\prod_{i=1}^n X_i\right)^{\theta-1}$$

   with log-likelihood

$$\ell(\theta) = n \log\theta + (\theta - 1)\sum_{i=1}^n \log(X_i).$$

   The critical points are at

$$\frac{\partial \ell}{\partial \theta} = \frac{n}{\theta} + \sum_{i=1}^n \log(X_i) \iff \theta = \frac{-n}{\sum_{i=1}^n \log(X_i)}.$$

   This is indeed a maximum; we have

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{n}{\theta^2} < 0$$

   Since the maximum for ℓ occurs at the same place as for L (log is increasing), we conclude that

$$\hat{\theta}_{\text{MLE}} = \frac{-n}{\sum_{i=1}^n \log(X_i)},$$

   as desired. □

3. For each of the following three sets of 10 observations from the given distribution, calculate the values of the maximum likelihood estimate $\hat{\theta}$:

(i)	0.0256	0.3051	0.0278	0.8971	0.0739
	0.3191	0.7379	0.3671	0.9763	0.0102
(ii)	0.9960	0.3125	0.4374	0.7464	0.8278
	0.9518	0.9924	0.7112	0.2228	0.8609
(iii)	0.4698	0.3675	0.5991	0.9513	0.6049
	0.9917	0.1551	0.0710	0.2110	0.2154

Sol. Computing directly we find

$$\boxed{\hat{\theta}_{\text{MLE}} \approx \begin{cases} 0.549 & \text{in case } (i) \\ 2.210 & \text{in case } (ii) \\ 0.959 & \end{cases}}$$

Exercise 7.3 (based on HTZ, Exercise 6.4.5 and 6.6.4)**.** Let $X_1, X_2, \ldots X_n \overset{i.i.d.}{\sim} F_\theta$, where F_θ has probability density function

$$f_\theta(x) = \frac{1}{\theta^2} x e^{-x/\theta}, \qquad 0 < x < \infty$$

where θ is an unknown parameter with possible values $0 < \theta < \infty$.

1. Determine the maximum likelihood estimator, $\hat{\theta}$.

   Sol. The likelihood function is given by

$$L(\theta) = \prod_{i=1}^n f_\theta(X_i) = \prod_{i=1}^n \frac{1}{\theta^2} X_i e^{-X_i/\theta} = \frac{1}{\theta^{2n}} \left(\prod_{i=1}^n X_i \right) e^{-\frac{1}{\theta}\sum_{i=1}^n X_i}$$

   and so the log-likelihood reads

$$\ell(\theta) = -2n\log(\theta) + \sum_{i=1}^n \log(X_i) - \frac{1}{\theta}\sum_{i=1}^n X_i.$$

   Differentiating, the critical points are at

$$\frac{\partial \ell}{\partial \theta} = -\frac{2n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^n X_i = 0 \iff \boxed{\hat{\theta} = \frac{1}{2n}\sum_{i=1}^n X_i}.$$

   This is indeed a maximum; we have

$$\frac{\partial^2 \ell}{\partial \theta^2} = \frac{2n}{\theta^2} - \frac{2}{\theta^3}\sum_{i=1}^n X_i =$$

   which at $\hat{\theta}$ equals

$$\frac{\partial^2 \ell}{\partial \theta^2}\left(\hat{\theta}\right) = \frac{2n}{\hat{\theta}^2} - \frac{2}{\hat{\theta}^3}\left(2n\hat{\theta}\right) = \frac{2n}{\hat{\theta}^2} - \frac{4n}{\hat{\theta}^2} = -\frac{2n}{\hat{\theta}^2} < 0$$

   and so the second derivative test tells us that $\hat{\theta}$ is indeed the MLE.

2. Compute $\text{bias}(\hat{\theta}, \theta)$.

   Sol. The bias is given by

$$\text{bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta}] - \theta = \frac{1}{2n}\sum_{i=1}^n \mathbb{E}[X_i] - \theta.$$

We have to compute $\mathbb{E}[X_i]$ directly. We have

$$\mathbb{E}[X_i] = \int_0^\infty x \cdot \frac{1}{\theta^2} x e^{-x/\theta} dx = \frac{1}{\theta^2} \int_0^\infty x^2 e^{-x/\theta} dx.$$

Using a u-substiuttion $u = \frac{x}{\theta}$ and then integrating by parts, we find

$$\begin{aligned}\frac{1}{\theta^2} \int_0^\infty x^2 e^{-x/\theta} dx &= \theta \int_0^\infty u^2 e^{-u} du = \theta \left(-u^2 e^{-u} \Big|_0^\infty + \int_0^\infty 2ue^{-u} du \right) \\ &= \theta \left(-u^2 e^{-u} \Big|_0^\infty - 2ue^{-u} \Big|_0^\infty + \int_0^\infty 2e^{-u} du \right) = \theta \left(-u^2 e^{-u} \Big|_0^\infty - 2ue^{-u} \Big|_0^\infty - 2e^{-u} \Big|_0^\infty \right) = 2\theta.\end{aligned}$$

So, $\mathbb{E}[X_i] = 2\theta$ and

$$\text{bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta}] - \theta = \frac{1}{2n} \sum_{i=1}^n \mathbb{E}[X_i] - \theta = \frac{1}{2n} \sum_{i=1}^n (2\theta) - \theta = \frac{2n\theta}{2n} - \theta = \theta - \theta = \boxed{0}.$$

3. Determine $\text{MSE}(\hat{\theta}, \theta)$.

Sol. Since the estimator is unbiased, $\text{MSE}(\hat{\theta}, \theta) = \text{Var}\left(\hat{\theta}\right)$, which we compute as

$$\text{Var}\left(\hat{\theta}\right) = \text{Var}\left(\frac{1}{2n} \sum_{i=1}^n X_i \right) = \frac{1}{4n^2} \sum_{i=1}^n \text{Var}(X_i).$$

Now, $\mathbb{E}[X_i] = 2\theta$ so

$$\text{Var}(X_i) = \mathbb{E}\left[(X_i - 2\theta)^2 \right] = \frac{1}{\theta^2} \int_0^\infty (x - 2\theta)^2 x e^{-\frac{x}{\theta}} \, dx = \int_0^\infty \left(\frac{x}{\theta} - 2 \right)^2 x e^{-\frac{x}{\theta}} dx.$$

Using a $u - substitution$ $u = \frac{x}{\theta}$, we find

$$\text{Var}(X_i) = \theta^2 \int_0^\infty (u-2)^2 u e^{-u} \, du = \theta^2 \int_0^\infty \left(u^3 - 4u^2 + 4u \right) e^{-u} du.$$

The integration by parts above showed that

$$\int_0^\infty u e^{-u} \, du = 1, \qquad \int_0^\infty u^2 e^{-u} du = 2$$

and so we can also compute

$$\int_0^\infty u^3 e^{-u} du = -u^3 e^{-u} \Big|_0^\infty + 3 \int_0^\infty u^2 e^{-u} du = 6.$$

Thus,

$$\text{Var}(X_i) = \theta^2 \int_0^\infty \left(u^3 - 4u^2 + 4u \right) e^{-u} du = \theta^2 (6 - 8 + 4) = 2\theta^2$$

and thus

$$\text{MSE}(\hat{\theta}, \theta) = \frac{1}{4n^2} \sum_{i=1}^n (2\theta^2) = \frac{2n\theta^2}{4n^2} = \boxed{\frac{\theta^2}{2n}}.$$

4. Show that $\hat{\theta}$ is consistent for θ.

Proof. Using the law of large numbers, we have

$$\frac{1}{2n} \sum_{i=1}^n X_i = \frac{1}{n} \sum_{i=1}^n \frac{X_i}{2} \xrightarrow{p} \mathbb{E}\left[\frac{X_i}{2} \right] = \frac{2\theta}{2} = \theta,$$

establishing the desired consistency. □

Alternatively, we could also use the MSE to show that

$$\mathbb{P}\left(|\hat{\theta} - \theta| \geq \epsilon\right) \leq \frac{\mathrm{MSE}(\hat{\theta}, \theta)}{\epsilon^2} = \frac{\theta^2}{2n\epsilon^2} \to 0$$

for any $\epsilon > 0$, and so $\hat{\theta} \xrightarrow{p} \theta$, establishing consistency.

5. Determine a constant σ such that

$$\frac{\sqrt{n}}{\sigma}\left(\hat{\theta} - \theta\right) \xrightarrow{d} \mathcal{N}(0,1).$$

Sol. Note that

$$\hat{\theta} = \frac{1}{2n}\sum_{i=1}^{n} X_i = \frac{1}{n}\sum_{i=1}^{n} \frac{X_i}{2}.$$

The random variables $\frac{X_i}{2}$ are i.i.d., and our computations above showed that

$$\mathbb{E}\left[\frac{X_i}{2}\right] = \frac{1}{2}\mathbb{E}[X_i] = \frac{2\theta}{2} = \theta, \qquad \mathrm{Var}\left(\frac{X_i}{2}\right) = \frac{1}{4}\mathrm{Var}(X_i) = \frac{2\theta^2}{4} = \frac{\theta^2}{2}.$$

So, by the standard CLT,

$$\frac{\sqrt{n}}{\sqrt{\frac{\theta^2}{2}}}\left(\hat{\theta} - \theta\right) \xrightarrow{d} \mathcal{N}(0,1)$$

so $\boxed{\sigma = \frac{\theta}{\sqrt{2}}}$ gives the desired convergence.

6. Compute the Fisher information, $I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f_{X_1;\theta}(X_1)\right)^2\right]$.

Sol. Notice that

$$\log f_{X_1;\theta}(x) = -2\log\theta + \log x - \frac{x}{\theta}$$

so

$$\frac{\partial}{\partial\theta}\log f_{X_1;\theta}(x) = \frac{-2}{\theta} + \frac{x}{\theta^2}$$

and

$$I(\theta) = \mathbb{E}\left[\left(\frac{X_1}{\theta^2} - \frac{2}{\theta}\right)^2\right] = \mathbb{E}\left[\frac{(X - 2\theta)^2}{\theta^4}\right] = \frac{1}{\theta^4}\mathrm{Var}(X_1) = \frac{2\theta^2}{\theta^4} = \boxed{\frac{2}{\theta^2}}.$$

7. Does $\hat{\theta}$ attain the Cramér-Rao lower bound?

Sol. For any unbiased estimator T, we have

$$\mathrm{Var}(T) \geq \frac{1}{nI(\theta)} = \frac{\theta^2}{2n}.$$

The quantity on the right, the Cramér-Rao lower bound, is precisely the variance of $\hat{\theta}$, so $\boxed{\text{yes}}$ it does.